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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

CV2505-1 – SUSTAINABILITY ENGINEERING

(Open Elective)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. Which tool is commonly used in systems thinking?
A) Flowcharts
B) SWOT analysis
C) Life cycle analysis
D) Stakeholder analysis
2. What does life cycle thinking focus on?
A) Production phase only
B) Recycling phase only
C) Entire product lifecycle from cradle to grave
D) Distribution phase only
3. Life Cycle Assessment (LCA) does NOT include:
A) Manufacturing
B) Raw material extraction
C) End-of-life disposal
D) Future financial profit estimation
4. Which sustainable practice reduces waste in production?
A) Over-extraction
B) Efficient resource utilization
C) Burning of excess materials
D) Increasing landfill use
5. A key benefit of green buildings is:
A) Reduced lifespan
B) Improved indoor air quality
C) Increased operational costs
D) Higher environmental degradation
6. Sustainable cities rely heavily on:
A) Urban green spaces
B) Fossil fuel consumption
C) High waste generation
D) Heavy industrialization
7. Which building is an example of a green building?
A) Burj Khalifa
B) The Edge, Amsterdam
C) Empire State Building
D) Beijing National Stadium
8. Which rating system focuses on water efficiency?
A) ENERGY STAR
B) GRIHA
C) BREEAM
D) LEED
9. What is the main goal of designing for sustainability?
A) Minimizing costs
B) Maximizing environmental, social, and economic benefits
C) Increasing resource consumption
D) Enhancing aesthetics
10. What is the purpose of emission regulations in sustainability?
A) To increase air pollution
B) To promote fossil fuel use
C) To reduce greenhouse gas emissions
D) To allow unlimited emissions
11. What does "urban mining" refer to in e-waste management?
A) Extracting valuable materials from discarded electronics
B) Digging for metals in landfills
C) Building mines in urban areas
D) Extracting non-renewable resources

12. What is the aim of disaster mitigation?
 A) To eliminate disaster response plans
 B) To increase disaster impact
 C) To delay recovery processes
 D) To minimize the risks and effects of disasters
13. What does an LCA for buildings assess?
 A) Aesthetic features of the design
 B) Environmental impacts over the building's entire lifecycle
 C) Time required for construction
 D) Cost of materials only
14. What does carbon footprint measure?
 A) Total carbon dioxide equivalent emissions
 B) Raw material cost
 C) Water usage during production
 D) Physical size of the product
15. Which LCA methodology focuses on long-term sustainability?
 A) Cradle-to-gate
 B) Gate-to-gate
 C) Cradle-to-cradle
 D) Gate-to-grave
16. What is a primary challenge in sustainable engineering design?
 A) Low customer demand
 B) Unlimited budget constraints
 C) High resource availability
 D) Lack of clear regulations
17. What does the term "energy intensity" measure?
 A) Speed of energy production
 B) Total cost of energy in a system
 C) Thermal efficiency of a process
 D) Energy consumption per unit output
18. Which city is an example of sustainable urban planning?
 A) Copenhagen, Denmark
 B) Beijing, China
 C) Las Vegas, USA
 D) Dubai, UAE
19. What is one lesson learned from failures in sustainable projects?
 A) Conventional methods outperform sustainable processes.
 B) Lack of environmental metrics has no impact.
 C) Ignoring stakeholder engagement leads to project failure.
 D) Sustainability eliminates the need for cost analysis.
20. What is an important criterion in selecting materials for sustainable design?
 A) High extraction costs
 B) Limited usability
 C) Low energy efficiency
 D) Durability and recyclability

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Define sustainable engineering and compare it with traditional engineering approaches to highlight the differences.	08	L*2	1	2,7
	b) Explain the role of recycled materials in green building and describe how they contribute to sustainability.	08	L2	2	2
2.	a) Discuss how engineers can contribute to achieving the United Nations Sustainable Development Goals (SDGs) with relevant examples.	08	L2	1	1,7
	b) Explain the different levels of LEED certification and outline the criteria required to achieve them.	08	L2	2	1,2
3.	a) Explain the concept of <i>green engineering</i> and illustrate its application with a suitable example.	08	L2	1	2,7
	b) Discuss the challenges encountered during the implementation of a sustainable transport system and interpret how they were overcome.	08	L2	2	1,2
Unit – II					
4.	a) Explain how 3D printing can contribute to green manufacturing and illustrate its sustainable benefits.	08	L2	3	2.4

b)	Explain how uncertainties in LCA results can be addressed during the interpretation stage.	08	L2	4	1,7
5. a)	Define a green data centre and describe its key features that promote sustainability.	08	L2	3	1,2
b)	Compare Environmental Life Cycle Costing (LCC) with traditional cost analysis, highlighting the key differences.	08	L2	4	1,2
6. a)	Explain what disaster mitigation strategies are and discuss their importance in engineering practices.	08	L2	3	1,2
b)	Discuss the challenges involved in integrating LCC with LCA in environmental assessments.	08	L2	4	2
Unit – III					
7. a)	Discuss the common challenges encountered in preparing a sustainability report and explain the lessons that can be learned from them.	08	L2	5	1,7
b)	Explain how the transition from conventional to sustainable engineering design impacts the overall design process.	08	L2	5	2
8. a)	Discuss a real-world example of a sustainable project in a specific engineering discipline and highlight the sustainable practices adopted.	08	L2	5	2,7
b)	Describe the essential components of a sustainability report and interpret their significance.	08	L2	5	2,7

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
